

Google Interview Handbook

1. Company Overview

Google is a global leader in technology, specializing in internet-related services and products. Known for its innovative culture and “Googliness,” it remains one of the most sought-after employers for software engineers.

2. Hiring Process (2025-2026)

The Google interview process typically takes 8-12 weeks and includes:

- **Recruiter Screen:** 30-minute call focusing on background and role fit.
- **Technical Phone Screen:** 45-60 minutes on Google Docs, focusing on DSA.
- **Onsite Interviews:** 4-5 rounds (45-60 mins each):
 - **Coding Rounds (2-3):** Data Structures and Algorithms.
 - **System Design (1-2):** For mid/senior roles (L4+).
 - **Googliness & Leadership (1):** Behavioral questions and cultural fit.
- **Team Matching:** Post-onsite process to find the right team.

3. Data Structures & Algorithms (DSA)

Frequently Asked Problems

Topic	Problem Example	Difficulty
Graphs	Word Ladder, Course Schedule	Medium/Hard
Dynamic Programming	Longest Palindromic Substring	Medium/Hard
Trees	Binary Tree Maximum Path Sum	Hard
Arrays/Strings	Median of Two Sorted Arrays	Hard

Expected Approach

- **Clarify:** Ask questions to define constraints.

- **Brute Force:** Start with a simple solution.
- **Optimize:** Aim for optimal Time/Space complexity.
- **Code:** Write clean, modular code on Google Docs.
- **Test:** Dry run with edge cases.

4. Machine Learning (ML) & AI System Design

ML Theory & Questions

- **Core Concepts:** Bias-Variance tradeoff, Overfitting/Underfitting, Regularization (L1/L2).
- **Deep Learning:** Transformers architecture, Attention mechanism, CNN/RNN fundamentals.
- **Generative AI:** LLM fine-tuning, RAG (Retrieval-Augmented Generation) architecture, Prompt Engineering.

AI System Design

- **Recommendation Systems:** Design YouTube's video recommendation engine.
- **Search Ranking:** Design Google Search ranking for news articles.
- **Ads Ranking:** Designing a real-time bidding system for ads.

5. System Design (High-Level & Low-Level)

- **High-Level Design (HLD):** Scalability, Load Balancing, Caching (Redis), Database Sharding, Microservices.
- **Low-Level Design (LLD):** Object-Oriented Design (OOD), Design Patterns (Singleton, Factory, Observer).
- **Common Questions:** Design a URL Shortener (TinyURL), Design Google Drive, Design a Web Crawler.

6. Behavioral & Googliness

Key Themes

- **Leadership:** Leading without authority.
- **Ambiguity:** Handling projects with unclear requirements.

- **Collaboration:** Resolving conflicts with teammates.
- **Ownership:** Taking responsibility for failures.

Example Questions

- “Tell me about a time you had a conflict with a teammate.”
- “Describe a situation where you had to make a decision with incomplete information.”
- “Why do you want to work at Google?”

7. Online Assessment (OA)

- **Platform:** HackerRank or internal tool.
- **Format:** 2-3 coding problems in 60-90 minutes.
- **Difficulty:** LeetCode Medium/Hard.

8. Preparation Tips

- Practice coding on a plain text editor (Google Docs).
- Focus on “why” and trade-offs in System Design.
- Use the STAR method for behavioral questions.

9. References

- [Google Careers - How We Hire](#)
- [I Got An Offer - Google Interview Questions](#)
- [LeetCode Discuss - Google Interview Experiences](#)